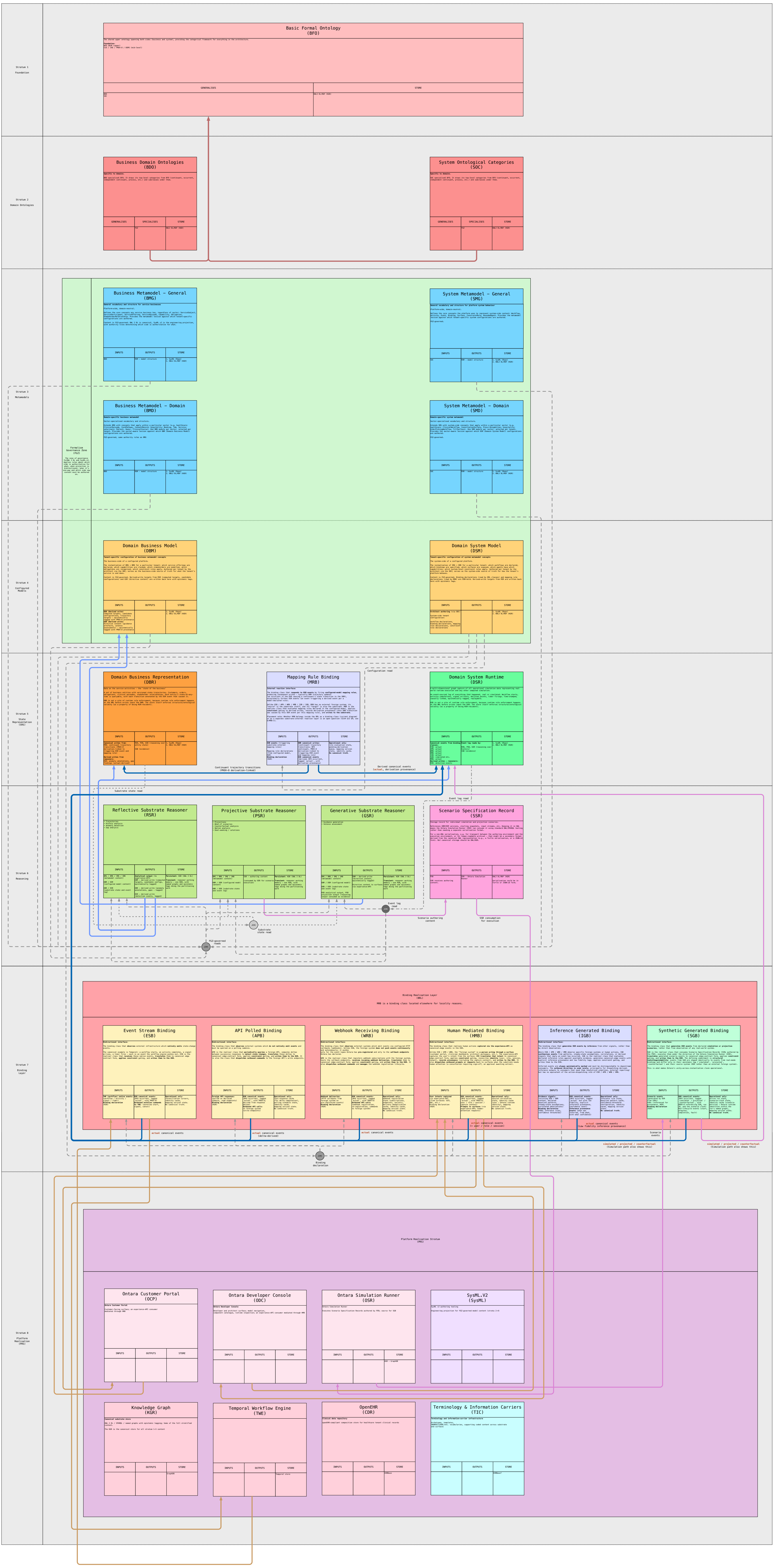


Stratum 1

Foundation

## Basic Formal Ontology
## (BFO)

The shared upper ontology spanning both sides (business and system), providing the categorical framework for everything in the architecture.

**Foundation:**
BFO 2020 (upper)
CCO / IAO / PROV-O / OGMS (mid-level)

| GENERALISES | STORE |
|---|---|
| BDO<br>SOC | OWL2 DL/RDF (KGR) |

Stratum 2

Domain Ontologies

## Business Domain Ontologies
## (BDO)

**Specific to domains.**

BDO specialiseS BFO. It draws its top-level categories from BFO (continuant, occurrent, independent continuant, process, etc.) and subclasses under them.

| GENERALISES | SPECIALISES | STORE |
|---|---|---|
|  | FGZ | OWL2 DL/RDF (KGR) |

## System Ontological Categories
## (SOC)

**Specific to domains.**

SOC specialiseS BFO. It draws its top-level categories from BFO (continuant, occurrent, independent continuant, process, etc.) and subclasses under them.

| GENERALISES | SPECIALISES | STORE |
|---|---|---|
|  | FGZ | OWL2 DL/RDF (KGR) |

Stratum 3

Metamodels

Formalism
Governance Zone
(FGZ)

The zone of governance forOWL 2 DL and SysML.v2. Applies rules about which side is authoritative for what, when projection is bidirectional, when it's one-way and which side new content must be authored on.

## Business Metamodel – General
## (BMG)

**General vocabulary and structure for service businesses**

Platform-wide, domain-neutral.

Defines the core concepts any service business has, regardless of sector: ServiceSubject, ServiceParticipant, ServiceOffering, ServiceEpisode, Capability, Obligation, StakeholderRelationship. Provides the metamodel lexicon against which tenant-specific configurations are authored.

Content is FGZ-governed: OWL 2 DL is canonical, SysML v2 is the engineering projection, with authority rules determining which side is authoritative for what.

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| BDO | RSM – model structure | 1. SysML (Repo)<br>2. OWL2 DL/RDF (KGR) |

## System Metamodel – General
## (SMG)

**General vocabulary and structure for platform system behaviour**

Platform-wide, domain-neutral.

Defines the core concepts the platform uses to represent system-side content: Workflow, Activity, Event, Binding, Surface, ConstraintRule, BoundedAgent. Provides the metamodel lexicon against which tenant-specific system configurations are authored.

FGZ-governed.

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| SOC | RSM – model structure | 1. SysML (Repo)<br>2. OWL2 DL/RDF (KGR) |

## Business Metamodel – Domain
## (BMD)

**Domain-specific business metamodel**

Sector-specialised vocabulary and structure.

Extends BMG with concepts that apply within a particular sector (e.g. healthcare: ClinicalEpisode, CarePathway, ConsentRecord; hospitality: Booking, Tab, Service; veterinary: Patient, Owner, ClinicalCourse). One BMD module per sector, selected per tenant. Provides the sector-aware lexicon against which DBM (Domain Business Model) configurations are authored.

FGZ-governed, same authority rules as BMG

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| BDO | RSM – model structure | 1. SysML (Repo)<br>2. OWL2 DL/RDF (KGR) |

## System Metamodel – Domain
## (SMD)

**Domain-specific system metamodel**

Sector-specialised vocabulary and structure.

Extends SMG with system-side concepts that apply within a particular sector (e.g. healthcare: ClinicalWorkflow, ConsultationSurface, PrescribingAction; hospitality: OrderPickingWorkflow, TillSurface). One SMD module per sector, selected per tenant. Provides the sector-aware lexicon against which DSM (Domain System Model) configurations are authored.

FGZ-governed.

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| SOC | RSM – model structure | 1. SysML (Repo)<br>2. OWL2 DL/RDF (KGR) |

Stratum 4

Configured
Models

## Domain Business Model
## (DBM)

**Tenant-specific configuration of business metamodel concepts**

The business-side of a configured platform.

The instantiation of BMG + BMD for a particular tenant: which service offerings are declared, which capabilities are claimed, which stakeholders are modelled, which obligations are recognised, which constraint rules apply. Authored per tenant by the architect via the ODC; serves as the business-side source of truth for what the tenant's service is and does.

Content is FGZ-governed. Derived-write targets from RSR (computed targets, candidate configurations) and GSR (directive content) are written back here with epistemic tags.

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| **RSR (Derived write)**<br>Computed targets, candidate configurations, trajectory targets – epistemically tagged with PROV-O provenance<br>**GSR (Derived write)**<br>Directive content (guidance artefacts, valence assessments) – epistemically tagged with PROV-O provenance |  | 1. SysML (Repo)<br>2. OWL2 DL/RDF (KGR) |

## Domain System Model
## (DSM)

**Tenant-specific configuration of system metamodel concepts**

The system-side of a configured platform.

The instantiation of SMG + SMD for a particular tenant: which workflows are declared, which bindings are specified, which surfaces are exposed, which agents have which capabilities, which system-level constraint rules apply. Authored per tenant by the architect via the ODC; serves as the system-side source of truth for how the tenant's platform behaves.

Content is FGZ-governed. Binding declarations (read by BRL classes) and mapping rule declarations (read by MRB) are DSM-held. Derived-write targets from RSR are written back here with epistemic tags.

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| **Architect authoring** (via ODC)<br>System-side tenant configuration:<br>workflow declarations, binding declarations, mapping rule declarations, constraint rule declarations |  | 1. SysML (Repo)<br>2. OWL2 DL/RDF (KGR) |

Stratum 5

State
Representation
(SRS)

## Domain Business Representation
## (DBR)

Data on the service activities – the 'state of the business'

A set of business entities with versioned state trajectories. Customers, orders, obligations, clinical episodes, stakeholder relationships. Each entity's state-at-any-time is queryable, with each transition annotated by the DSR event that caused it.

DBR is not a site of runtime rule enforcement, because runtime rule enforcement happens at the BRL before writes reach the DBR. The store itself enforces structural/ontological validity (as a property of being KGR-resident).

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| **Canonical writes from:**<br>MRB: continuant trajectory transitions (PROV-O derivation-linked to triggering DSR event and mapping rule)<br>**Derived writes from reasoners:**<br>RSR: anomaly annotations, gap findings (actual-derived) | RSR, PSR, GSR (reasoning over entity state)<br>IGB (evidence) | 1. SysML (Repo)<br>2. OWL2 DL/RDF (KGR) |

## Mapping Rule Binding
## (MRB)

**Internal reaction interface:**

The binding class that **responds to DSR events** by firing **configured-model mapping rules**, producing consequent writes – typically DBR trajectory updates (an occurrent in the DSR causing a continuant's state transition in the DBR), occasionally further DSR events (an event triggering a derived event per a model-declared rule).

Unlike ESB / APB / WRB / HMB / IGB / SGB, MRB has no external foreign system; its "source" is the substrate itself, and its "target" is also the substrate. MRB is the realiser class that evaluates mapping rules declared in the configured model, applies **constraint** gating to derived writes, records derivation provenance (this DBR transition was caused by this DSR event per this mapping rule), and **writes to the substrate**.

Placement note: Whether MRB belongs inside the BRL as a binding class (current diagram) or as a separate substrate-internal reaction layer is an open question (S234 §12 Q4; see Q-MRB-1).

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| **DSR events** (triggering substrate-internal mapping rules)<br>Mapping rule declarations (from configured model, KGR)<br>**Binding declaration** (KGR) | **DBR canonical writes:** (continuant trajectory transitions; BFO continuant, PROV-O derivation-linked to triggering DSR event and mapping rule)<br>**DSR canonical events** (derived; BFO occurrent, tagged 'actual', with derivation provenance) | **Operational only:** rule evaluation state, pending-derivation queue, mapping version state, identity cache.<br>**No canonical truth.** |

## Domain System Runtime
## (DSR)

A multi-dimensional graph capture of all operational simulation data representing real world runtime execution and any other computed simulation.

An event-sourced log of everything that happened, real or simulated: Workflow starts, activity completions, signal arrivals, decision points, timer firings. Time-stamped, causally linked, epistemically tagged, replayable.

DSR is not a site of runtime rule enforcement, because runtime rule enforcement happens at the BRL before writes reach the DSR. The store itself enforces structural/ontological validity (as a property of being KGR-resident).

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| **Canonical events from binding classes:**<br>ESB: actual<br>APB: actual<br>WRB: actual<br>HMB: actual<br>IGB: actual<br>SGB: simulated etc.<br>MRB: actual<br>**Derived writes – reasoners:**<br>RSR: detection events | **Event log reads by:**<br>RSR, PSR, GSR (reasoning over history)<br>IGB (evidence) | 1. SysML (Repo)<br>2. OWL2 DL/RDF (KGR) |

Stratum 6

Reasoning

## Reflective Substrate Reasoner
## (RSR)

- Trajectories
- History analysis
- Anomaly detection
- Gap analysis

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| BMG / BMD / SMG / SMD (metamodel content)<br>DBM / DSM (configured model content)<br>DBR / DSR (substrate state and event log) | **Analytical output** (to experience-API)<br>DBM – derived write (computed targets, candidate configs, epistemically tagged)<br>DBR – derived write (anomaly annotations, gaps – tagged)<br>DSR – derived write (detection events, tagged) | **Persistent:** KGR (OWL 2 DL)<br>**Transient:** reasoner working memory – one KGR, with named graphs and epistemic tags doing the partitioning work |

## Projective Substrate Reasoner
## (PSR)

- Projections
- What-if scenarios
- Counterfactual analysis
- Option analysis
- Goal-seeking / solutions

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| BMG / BMD / SMG / SMD (metamodel content)<br>DBM / DSM (configured model content)<br>DBR / DSR (substrate state and event log) | SSR – authoring content<br>(consumed by OSR for scenario execution) | **Persistent:** KGR (OWL 2 DL)<br>**Transient:** reasoner working memory – one KGR, with named graphs and epistemic tags doing the partitioning work |

## Generative Substrate Reasoner
## (GSR)

- Guidance generation
- Valence assessment

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| BMG / BMD / SMG / SMD (metamodel content)<br>DBM / DSM (configured model)<br>DBR / DSR (substrate state and event log)<br>RSR analytical output, PSR projection output (reasoning output consumed as evidence) | DBM – derived write (directive content, epistemically tagged)<br>Directive content to surfaces via experience-API | **Persistent:** KGR (OWL 2 DL)<br>**Transient:** reasoner working memory – one KGR, with named graphs and epistemic tags doing the partitioning work |

## Scenario Specification Record
## (SSR)

Storage record for individual simulation and projection scenarios.

References DBM/DSM versions, starting snapshots, input streams, etc. Keeping it in OWL means the Ontara Simulation Runner (OSR) can consume it using standard OWL/SPARQL tooling rather than needing a separate serialisation format.

For a non-OWL serialisation, e.g. for transport between the authoring environment and the execution environment, or for human-readable archival – that might be a secondary format derived from the canonical OWL representation (e.g., a Turtle serialisation, or a JSON-LD form). But canonical storage should be OWL/KGR.

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| PSR<br>SSR receives authoring content. | OSR – Ontara Simulation Runner | OWL2 DL/RDF (KGR)<br>Serialisation could be in Turtle or JSON-LD form. |

Stratum 7

Binding
Layer

Binding Realisation Layer
(BRL)

MRB is a binding class located elsewhere for locality reasons.

## Event Stream Binding
## (ESB)

**Bidirectional interface:**

The binding class that **observes** external infrastructure which **natively emits** state-change events.

The canonical example is Temporal: a workflow starts, an activity completes, a signal arrives, a timer fires – each is an event the workflow engine pushes out. ESB is the realiser class that **receives** those native events, **translates** them to canonical edge contract form, **applies constraint** gating, and **writes them to the DSR.**

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| **TWE (workflow) native events:** (workflow / activity / signal / timer)<br>**Binding declaration** (KGR) | **DSR canonical events:** (BFO occurrent, tagged 'actual', with PROV-O)<br>**Outbound workflow commands to TWE:** (workflow start, signal, cancel) | **Operational only:** reconciliation cursors, identity cache, mapping version state.<br>No canonical truth. |

## API Polled Binding
## (APB)

**Bidirectional interface:**

The binding class that **observes** external systems which **do not natively emit events** and must be queried on a polling cadence.

APB is the realiser class that **periodically queries** a foreign API, computes deltas between successive responses to **detect state changes, translates** those deltas to canonical edge-contract form, applies **constraint** gating, and **writes them to the DSR.** It is also the class that **dispatches outbound commands** to the foreign API's write endpoints.

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| **Foreign API responses:** (polled on declared cadence, read endpoints)<br>**Binding declaration** (KGR) | **DSR canonical events:** (BFO occurrent, tagged 'actual', with PROV-O; derived from response deltas)<br>**Outbound API calls:** to foreign system (write endpoints) | **Operational only:** last-response cache (for delta computation), polling schedule state, identity cache, mapping version state.<br>No canonical truth. |

## Webhook Receiving Binding
## (WRB)

**Bidirectional interface:**

The binding class that **observes** external systems which emit events via configured HTTP callbacks (webhooks). Unlike ESB, the foreign system **does not push events continuously** and autonomously; it pushes only for the event types Ontara has **pre-registered** and only to the **callback endpoints** Ontara has declared.

WRB is the realiser class that registers webhook subscriptions with the foreign system, hosts the callback endpoints, **receives incoming webhook deliveries, translates** them to canonical edge-contract form, applies **constraint** gating, and **writes them to the DSR.** It also **dispatches outbound commands** and **manages** the webhook registration lifecycle.

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| **Webhook deliveries:** (HTTP callbacks from foreign system on pre-registered events)<br>**Binding declaration** (KGR) | **DSR canonical events:** (BFO occurrent, tagged 'actual', with PROV-O)<br>**Outbound API calls** (webhook registration, re-registration, commands to foreign system) | **Operational only:** webhook registration state (subscriptions, endpoints, secrets), delivery deduplication cache, identity cache, mapping version state.<br>No canonical truth. |

## Human Mediated Binding
## (HMB)

**Bidirectional interface:**

The binding class that realises human actions **captured via the experience-API** as canonical-edge events in the DSR.

Unlike ESB / APB / WRB, the "foreign system" is **a person acting through a surface** (customer portal, clinician dashboard, architect workspace, etc.). The experience-API captures the user's intent from the surface; HMB **translates that intent** to canonical edge-contract form, applies **constraint** gating, reconciles **identity** (user, role, session, tenant), **records provenance** (including the capture context), and **writes to the DSR.** It also **dispatches outbound prompts or requests** back to surfaces when the substrate needs human attention (a consultation requiring sign-off, an approval awaiting action).

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| **User intents captured** via experience-API (from surfaces: OCP, ODC, etc.)<br>Binding declaration (KGR) | **DSR canonical events:** (BFO occurrent, tagged 'actual', with PROV-O; user / role / session / capture context)<br>**Prompts to surfaces** (via experience-API; human-attention requests) | **Operational only:** session correlation state, identity cache (user → Ontara runtime identity), mapping version state.<br>No canonical truth. |

## Inference Generated Binding
## (IGB)

**Bidirectional interface:**

The binding class that **generates DSR events by inference** from other signals, rather than by direct observation.

Where ESB / APB / WRB / HMB observe specific foreign systems or human actions, IGB **synthesises events** from patterns, steady-state assumptions, correlations, or derived signals that imply an event has occurred. IGB is the realiser class that evaluates declared inference rules against available evidence, generates canonical-edge events with explicit inference provenance and low fidelity tags, applies constraint gating, and writes them to the DSR.

Inferred events are **always epistemically marked** and must be distinguishable to downstream consumers. The **outbound direction is used rarely,** principally for dispatching derived-inference outputs to consumers that need them (analytical pipelines, external reporting) but has no equivalent of the action-dispatching role of ESB / APB / WRB / HMB.

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| **Evidence signals:** (existing DSR / DBR content, declared baseline patterns, steady-state assumptions, correlated observations)<br>**Binding declaration** (KGR; inference rules, confidence thresholds) | **DSR canonical events:** (BFO occurrent, tagged 'actual' but with low fidelity, explicit inference provenance, confidence level)<br>**Inference provenance events** (what was inferred, from what, with what confidence) | **Operational only:** inference rule state, baseline pattern cache, confidence-threshold configuration, identity cache, mapping version state.<br>**No canonical truth.** |

## Synthetic Generated Binding
## (SGB)

**Bidirectional interface:**

The binding class that **generates DSR events** from declared **simulation or projection scenarios,** rather than from observation of any real-world system.

SGB is the realiser class that consumes Scenario Specification Records (SSRs authored by the PSR), executes them under the direction of the Ontara Simulation Runner (OSR), **translates** generated scenario events to canonical edge-contract form, applies **constraint** gating, and **writes them to the DSR.** The defining commitment is **structural indistinguishability:** events from SGB are shaped identically to events from real-mode bindings and differ only in their epistemic tag ('simulated', 'projected', 'counterfactual') and their source system (OSR rather than an external foreign system).

This is what makes Ontara's unity-across-instantiation claim operational.

| INPUTS | OUTPUTS | STORE |
|---|---|---|
| **Scenario events** (generated by OSR from SSRs)<br>SSRs (referenced for provenance, KGR)<br>**Binding declaration** (KGR) | **DSR canonical events:** (BFO occurrent, tagged 'simulated' / 'projected' / 'counterfactual', with PROV-O referencing SSR, run ID, and the scenario clock)<br>Run-lifecycle events (start, progress, completion, fault) | **Operational only:** active run state, scenario-clock state, identity cache (scenario entities → Ontara runtime identities within the run's named graph), mapping version state.<br>**No canonical truth.** |

Stratum 8

Platform
Realisation
(PRS)

Platform Realisation Stratum
(PRS)

## Ontara Customer Portal
## (OCP)

**Ontara Customer Portal**

Customer-facing surface; an experience-API consumer mediated through HMB

| INPUTS | OUTPUTS | STORE |
|---|---|---|
|  |  |  |

## Ontara Developer Console
## (ODC)

**Ontara Developer Console**

Developer and architect surface; model navigation, component catalogue, runtime inspection; an experience-API consumer mediated through HMB

| INPUTS | OUTPUTS | STORE |
|---|---|---|
|  |  |  |

## Ontara Simulation Runner
## (OSR)

Ontara Simulation Runner

Executes Scenario Specification Records authored by PSR; source for SGB

| INPUTS | OUTPUTS | STORE |
|---|---|---|
|  |  | KGR – GraphDB |

## SysML.V2
## (SysML)

SysML v2 authoring tooling

Engineering projection for FGZ-governed model content (strata 2–4)

| INPUTS | OUTPUTS | STORE |
|---|---|---|
|  |  |  |

## Knowledge Graph
## (KGR)

**Canonical substrate store**

OWL 2 DL / SPARQL / named graphs with epistemic tagging; home of the full stratified content.

The KGR is the canonical store for all stratum 1–5 content

| INPUTS | OUTPUTS | STORE |
|---|---|---|
|  |  | GraphDB |

## Temporal Workflow Engine
## (TWE)

| INPUTS | OUTPUTS | STORE |
|---|---|---|
|  |  | Temporal store |

## OpenEHR
## (CDR)

**Clinical data repository**

openEHR-compliant composition store for healthcare tenant clinical records

| INPUTS | OUTPUTS | STORE |
|---|---|---|
|  |  | EHRBase |

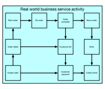
## Terminology & Information Carriers
## (TIC)

**Terminology and information-carrier infrastructure**

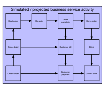
Archetypes, templates, SNOMED/LOINC/etc. vocabularies, supporting coded content across substrate and surfaces

| INPUTS | OUTPUTS | STORE |
|---|---|---|
|  |  | EHRBase? |

Ontara Platform
v3.2.0
20/04/2026



Real world business service activity



Simulated / projected business service activity